Did the Ads Actually Work?

Measuring what retargeting really causes

Sharath Sista · October 2026

Independent analysis of public data (Criteo Uplift v2.1, CC BY-NC-SA 4.0; Iowa Liquor Sales). Not affiliated with or endorsed by Criteo, Google or the State of Iowa.

THE QUESTION

Dashboards credit retargeting with sales. How many would have happened anyway?

How much do ads cause?

Not what they are credited with

Who is persuaded?

And who would buy anyway

Does it pay?

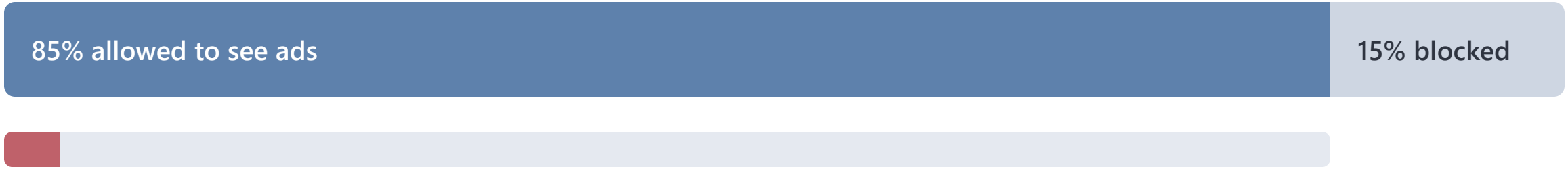
Profit per £1, not attributed return

How to measure next time?

When users can't be split

THE TEST

A real randomised test on 13,979,592 users



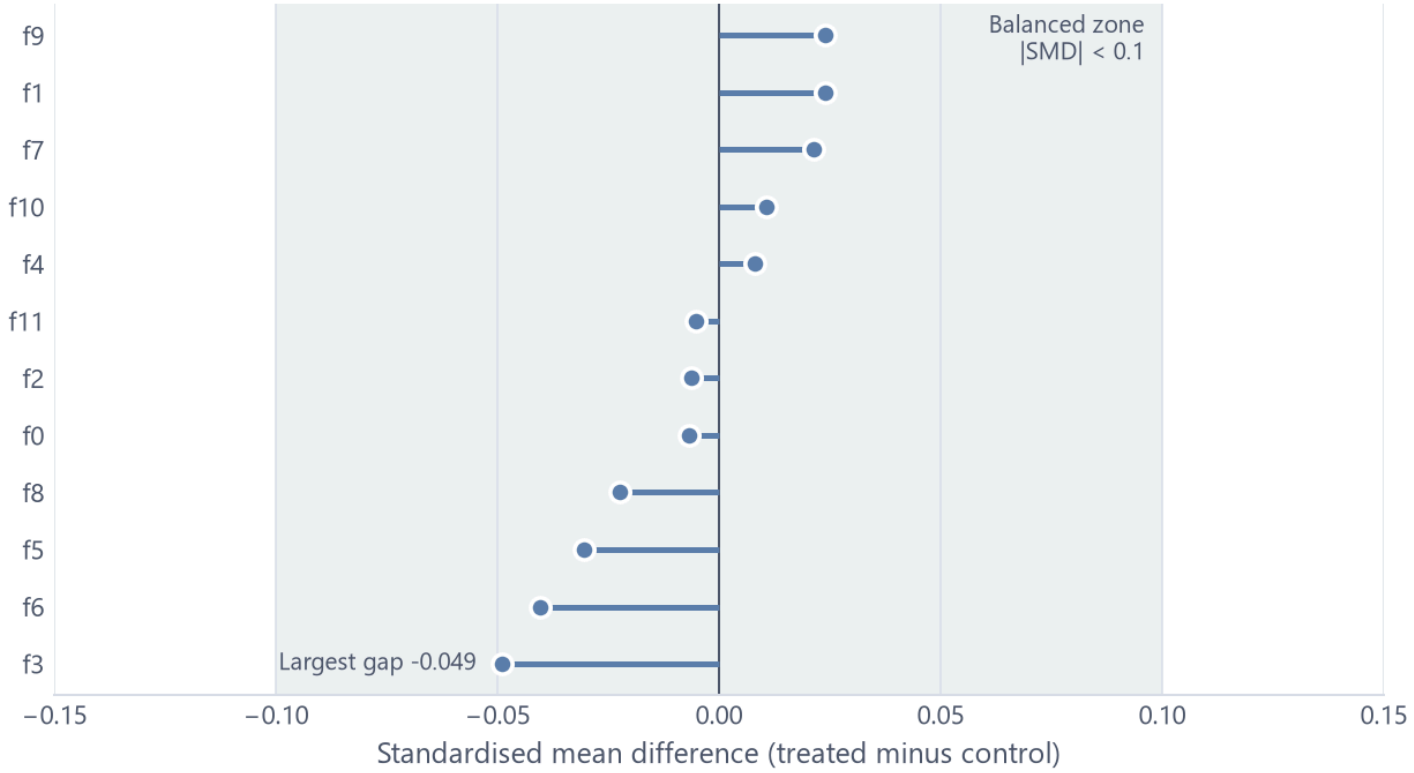
Only 3.6% of the allowed group actually saw an ad (red). Because the split is random, any difference between the groups was caused by the ads.

- Plus 2,700 simulated campaigns on real Iowa county sales to grade regional testing methods
- Criteo sub-sampled the data, so levels describe this file; the methods transfer

Source: Criteo Uplift v2.1 (Diemert et al.), CC BY-NC-SA 4.0 · levels describe the sub-sampled benchmark

Treated and control users look alike on all 12 features

Gap between group means in standard deviations, 13,979,592 users. Every gap is under half the 0.1 line.



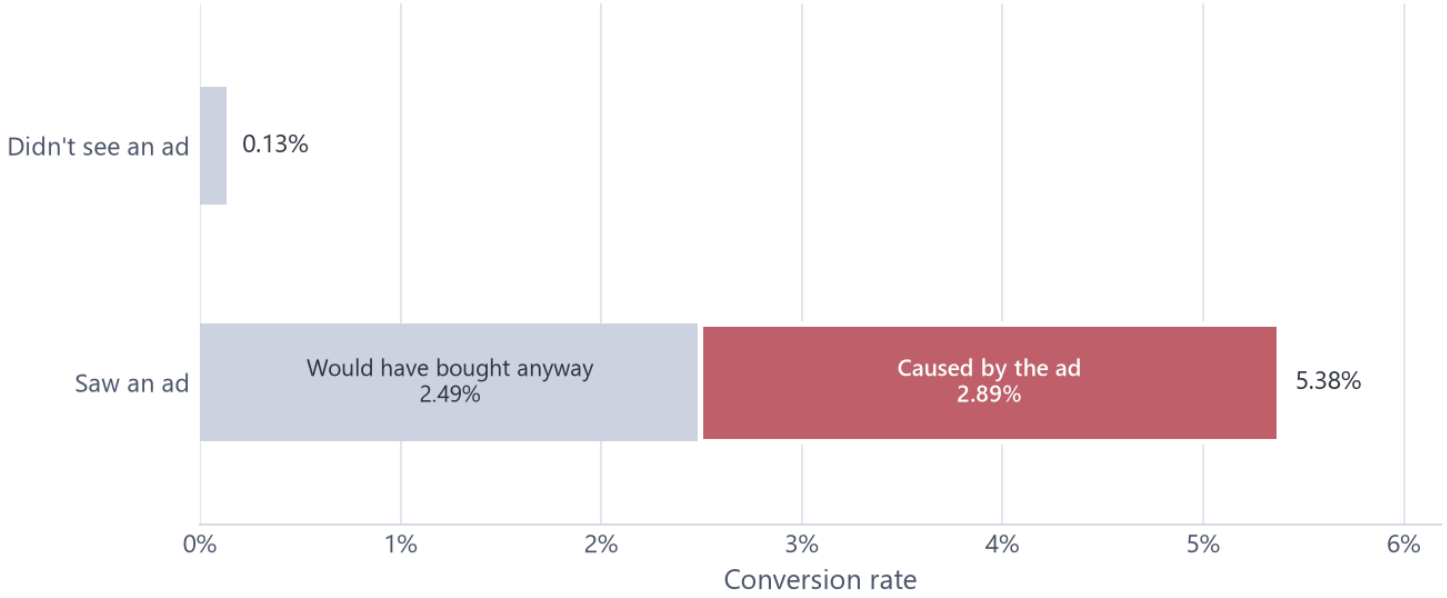
Source: Criteo Uplift v2.1 (Diemert et al., AdKDD 2018), randomised test on 13,979,592 users.

- Split exactly 85.00% treated, as designed
- A model tells the groups apart slightly: AUC 0.509 (placebo 0.499)
- Cause: several advertisers' tests merged; features were logged before any ad, so adjusting is valid
- It matters: visit lift +27% raw, +20% adjusted

Source: Criteo Uplift v2.1 (Diemert et al.), CC BY-NC-SA 4.0 · levels describe the sub-sampled benchmark

Seeing an ad looks like a 41x boost. The real effect is 2.2x

Users who saw an ad were already 19x more likely to buy. The test separates that from the ad.



Source: Criteo Uplift v2.1, 13,979,592 users. Rates describe this sub-sampled benchmark file, not Criteo's live results.

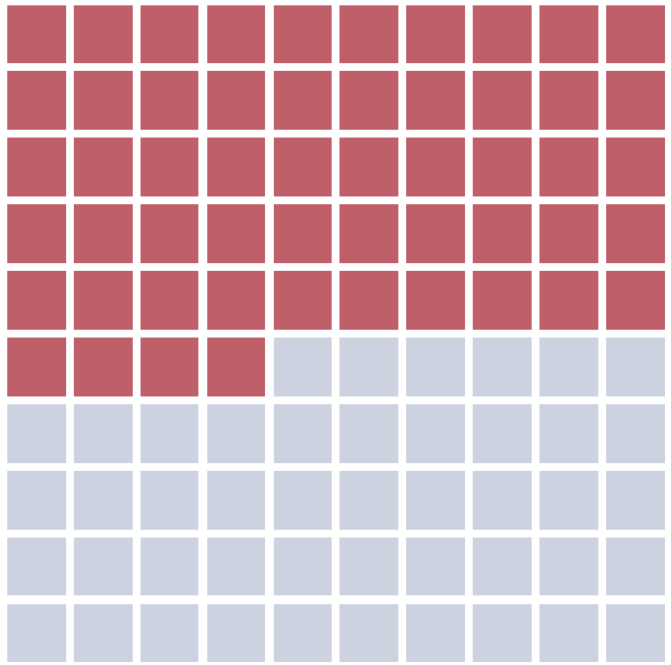
2.2x

real effect of seeing an ad

- Saw an ad vs didn't looks like 41x
- Those users were already 19x likelier to buy
- The real effect of seeing an ad: 2.2x (+2.89 points)

Only 54 of every 100 conversions credited to ads were caused by them

Conversions by users who saw an ad, split using the randomised test (doubly robust estimate).



54 caused by the ad

46 would have happened without the ad

95% interval: 50 to 58 caused. Attribution overstates the ads' conversions by about 1.9x. Without correcting the leak from finding 1, it would read 59.

Source: Criteo Uplift v2.1, 13,979,592 users. Shares describe this sub-sampled benchmark file.

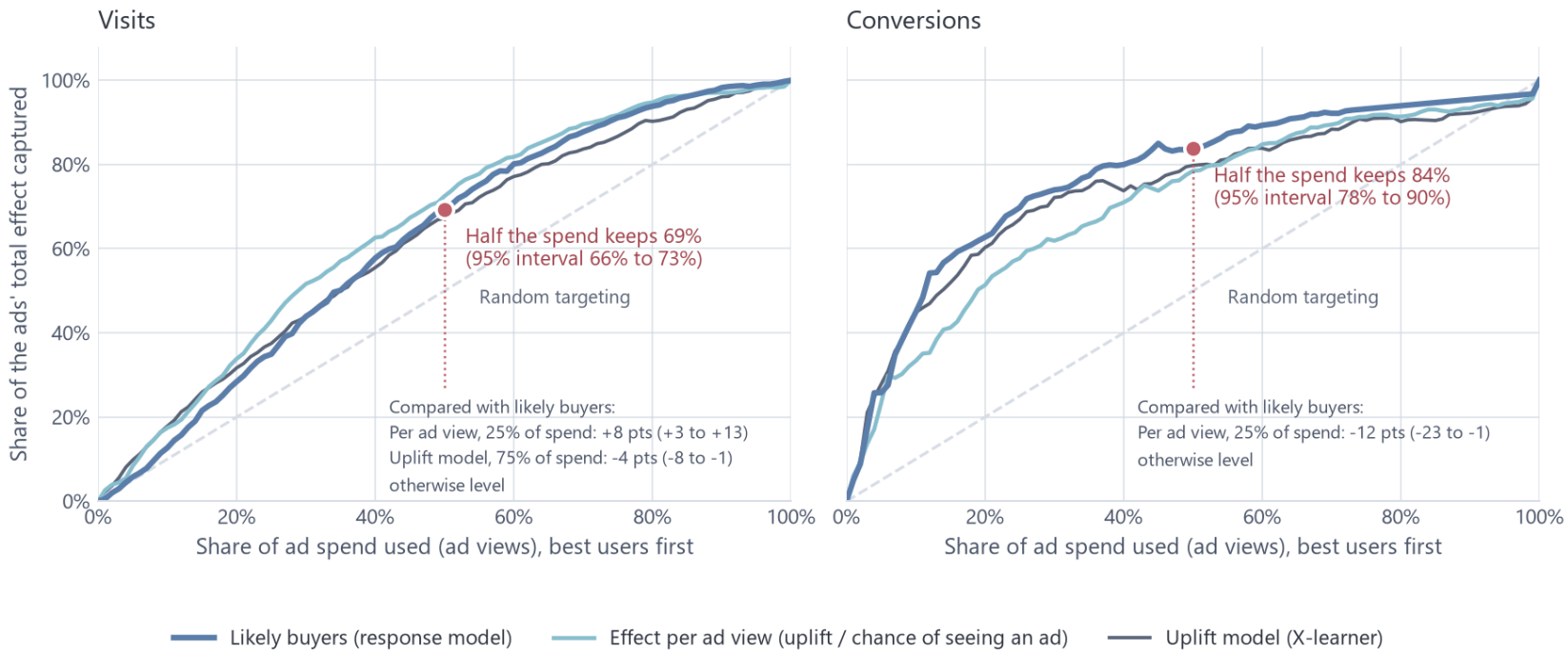
54

of every 100 credited conversions were caused by the ads

- 95% interval 50 to 58
- Attribution overstates retargeting's return by about 1.9x
- Ads work, but attribution also hands them the sales of keen shoppers

Half the ad spend, aimed at likely buyers, keeps 84% of the conversions ads cause

And 69% of the visits. Spend measured as ad views. Effect-per-ad-view ranking is exploratory (added after first results).



Source: Criteo Uplift v2.1, 13,979,592 users (80% train, 20% test). Shares describe this sub-sampled benchmark file.

- The top 20% of users hold 91% of the effect
- Half the spend, aimed at likely buyers, keeps 84% of conversions caused
- A simple likely-buyer model matched T- and X-learner uplift models on conversions

Each £1 of ads looks like £1.40 profit. Really it's £0.29, and half the spend loses money

Illustrative: assumes a dashboard ROAS of 8 and a 30% margin. Caused shares and the spend split come from the test.



Source: Criteo Uplift v2.1, 13,979,592 users; money figures use stated assumptions. Spend measured as ad views.

- Dashboard implies +£1.40 per £1; really +£0.29
- The weaker half of spend makes -£0.58 per £1
- Cutting it would roughly double profit (2.0x)

FINDING 7 · A RULE ANYONE CAN USE

At a 30% margin, retargeting needs a dashboard ROAS above 6.2 to pay

6.2
Break-even ROAS, all spend

3.7
Best half of spend

19.2
Weaker half of spend

Real ROAS = dashboard ROAS × **0.54**, the caused share from a holdout. No ROAS assumption needed.

Source: Criteo Uplift v2.1 (Diemert et al.), CC BY-NC-SA 4.0 · levels describe the sub-sampled benchmark

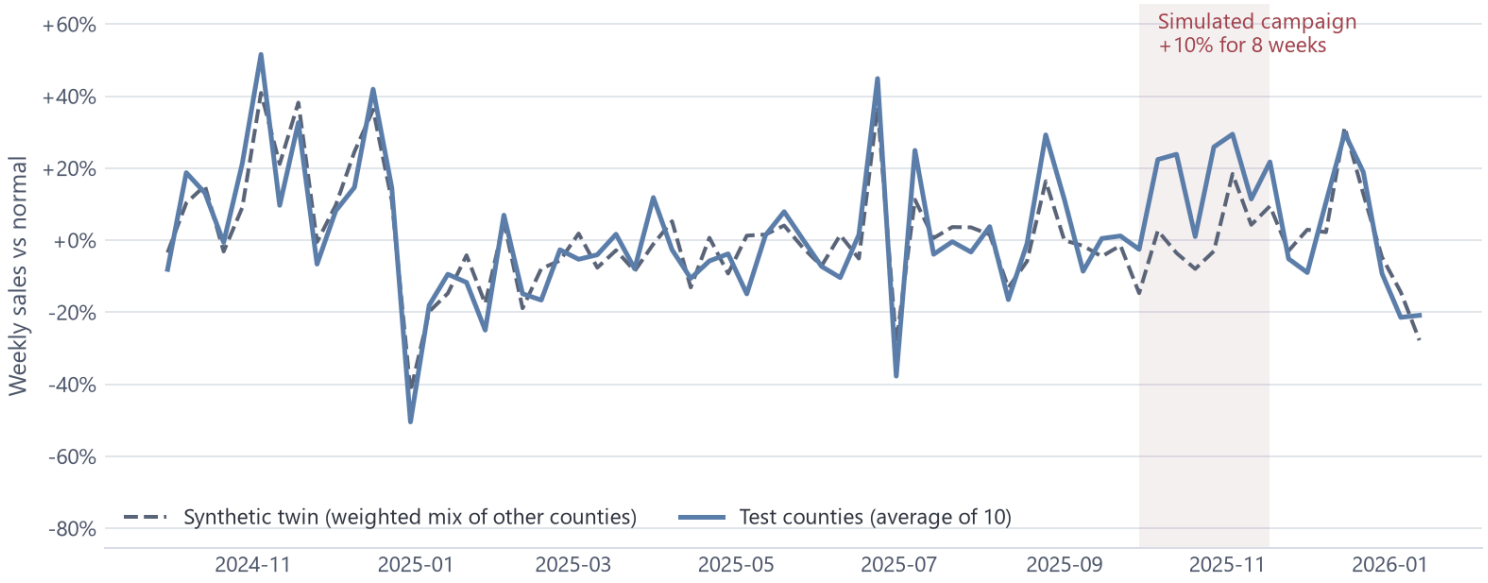
PART TWO

When you can't hold users out

TV, outdoor and some ad platforms can't split individual users. The fallback is a geo test.

Synthetic control estimates the +10% campaign at +16% (+6% to +26%)

Methods demonstration: real Iowa liquor sales, simulated campaign. Weekly sales swing a lot, so the interval is wide.

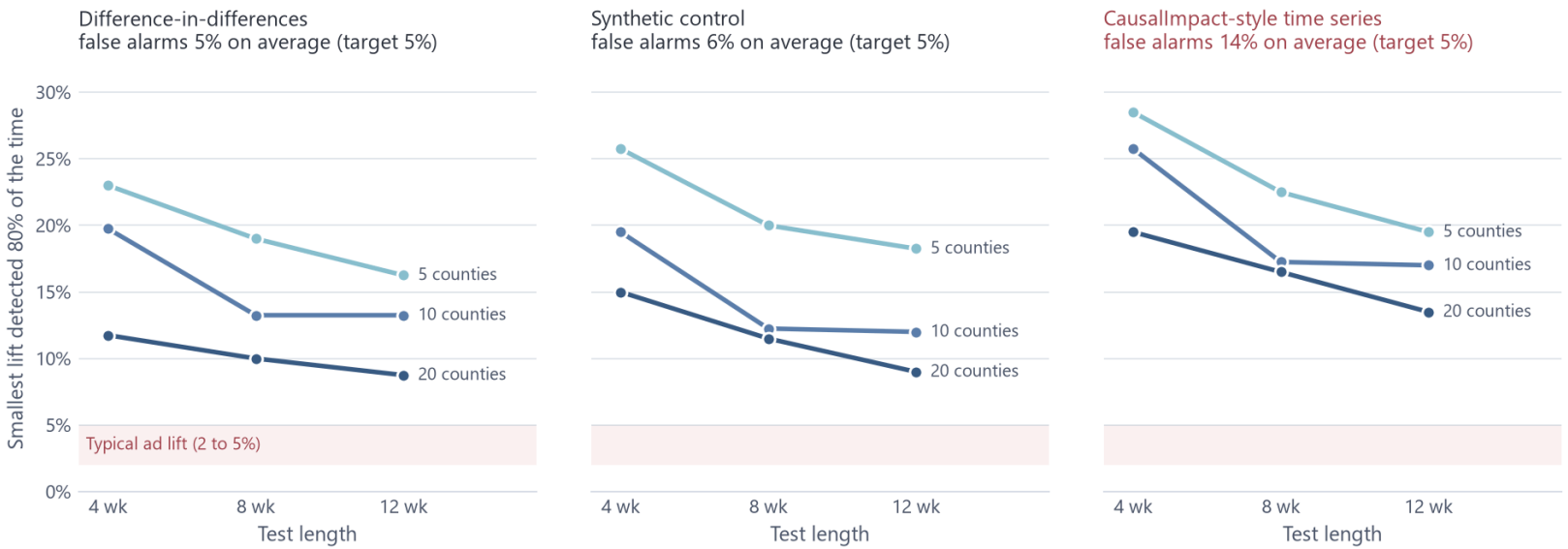


Source: Iowa Liquor Sales (BigQuery public data). Real sales, simulated campaign.

- A weighted mix of other counties (the synthetic twin) tracks the test counties
- During the campaign, the gap is the effect
- The campaign is invented, so the true answer is known

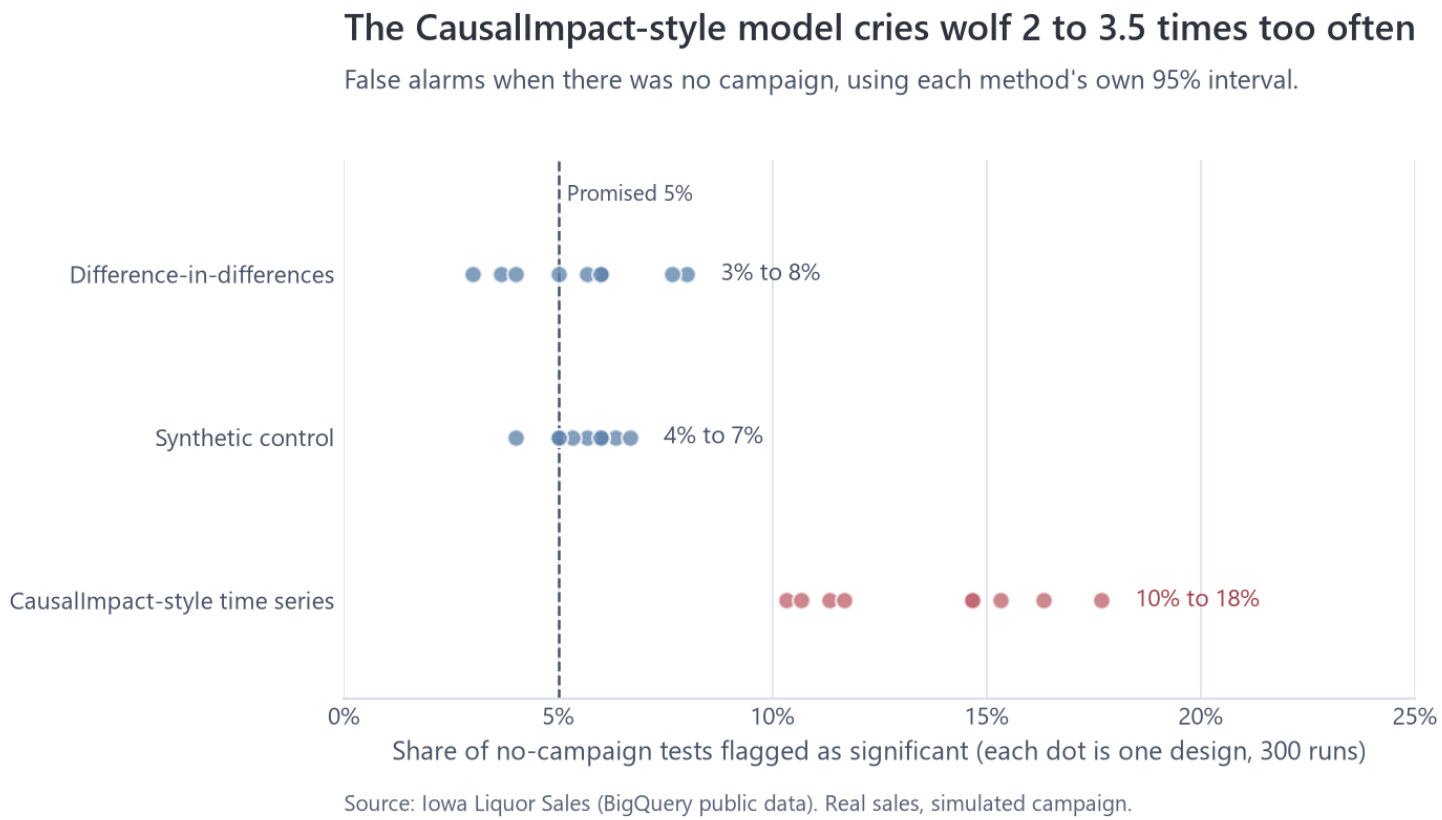
Even 20 counties for 12 weeks only reliably detect a lift of about 9%

Methods demonstration: real Iowa liquor sales, simulated campaigns. 300 runs per design, all methods held to a 5% false-alarm rate.



Source: Iowa Liquor Sales (BigQuery public data). Real sales, simulated campaign.

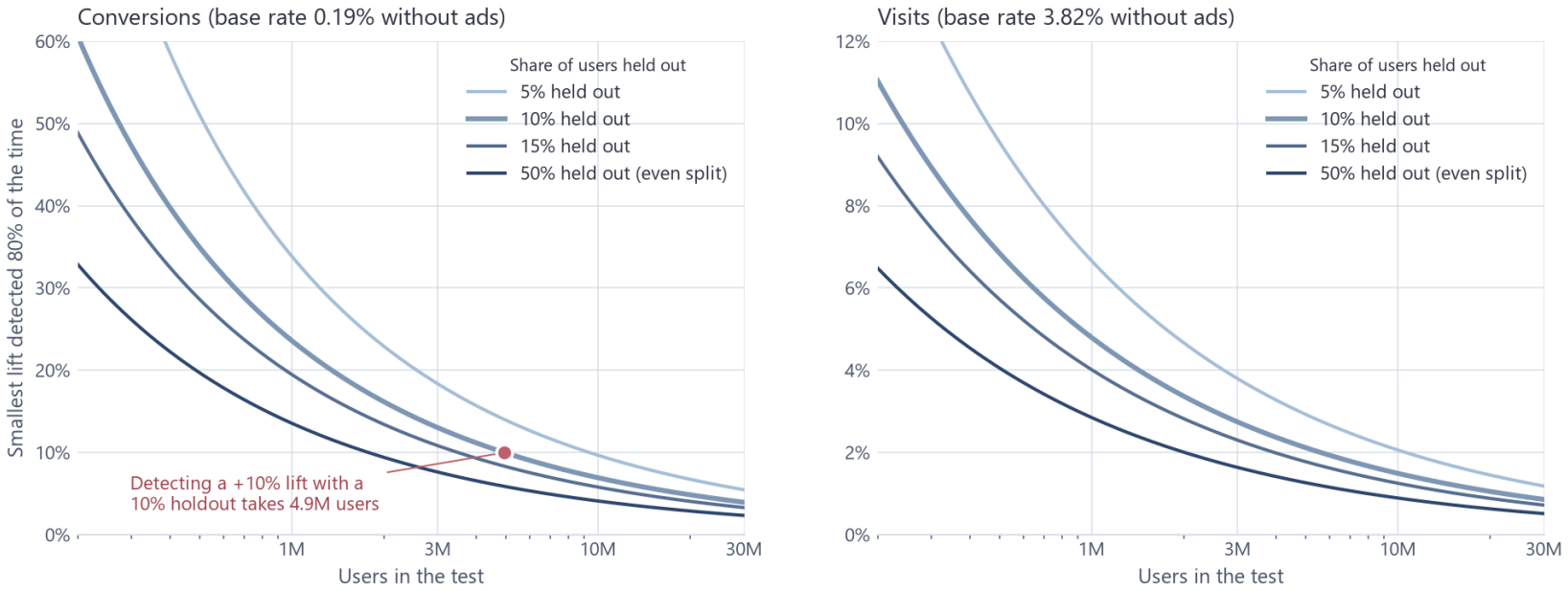
- Best design (20 counties, 12 weeks) reliably detects about 9%
- Typical ad effects are a few percent
- More regions help faster than more weeks; size the test by simulation first



- Difference-in-differences: 3% to 8% false alarms
- Synthetic control: 4% to 7%
- CausalImpact-style: 10% to 18%, promised 5%
- Placebo-test any model before trusting its intervals

Rare conversions need big tests: 5 million users to see a 10% lift with a 10% holdout

Smallest detectable lift by test size and share of users held out (5% false-alarm rate, 80% power). Visits need about 20x fewer users.



Base rates: Criteo Uplift v2.1 control group. Formula checked against the Criteo test's own precision (within 0.2 points).

- About 4.9 million users to see a +10% conversion lift with a 10% holdout
- 1.8 million with a 50/50 split
- Visits read early with about 0.9 million

THE PLAN

Measure what ads cause, not what they're credited with

1	**Report incremental ROAS next to attributed ROAS**	Only 54 of 100 credited conversions were caused
2	**Keep an always-on, properly sized holdout**	4.9 million users for a +10% conversion lift
3	**Test the budget cut before making it**	Half the spend keeps 84%; the weaker half loses money
4	**Start with the simple targeting model**	It matched uplift models on conversions
5	**Geo tests only where holdouts are impossible**	Best design detects about 9%; placebo-check models

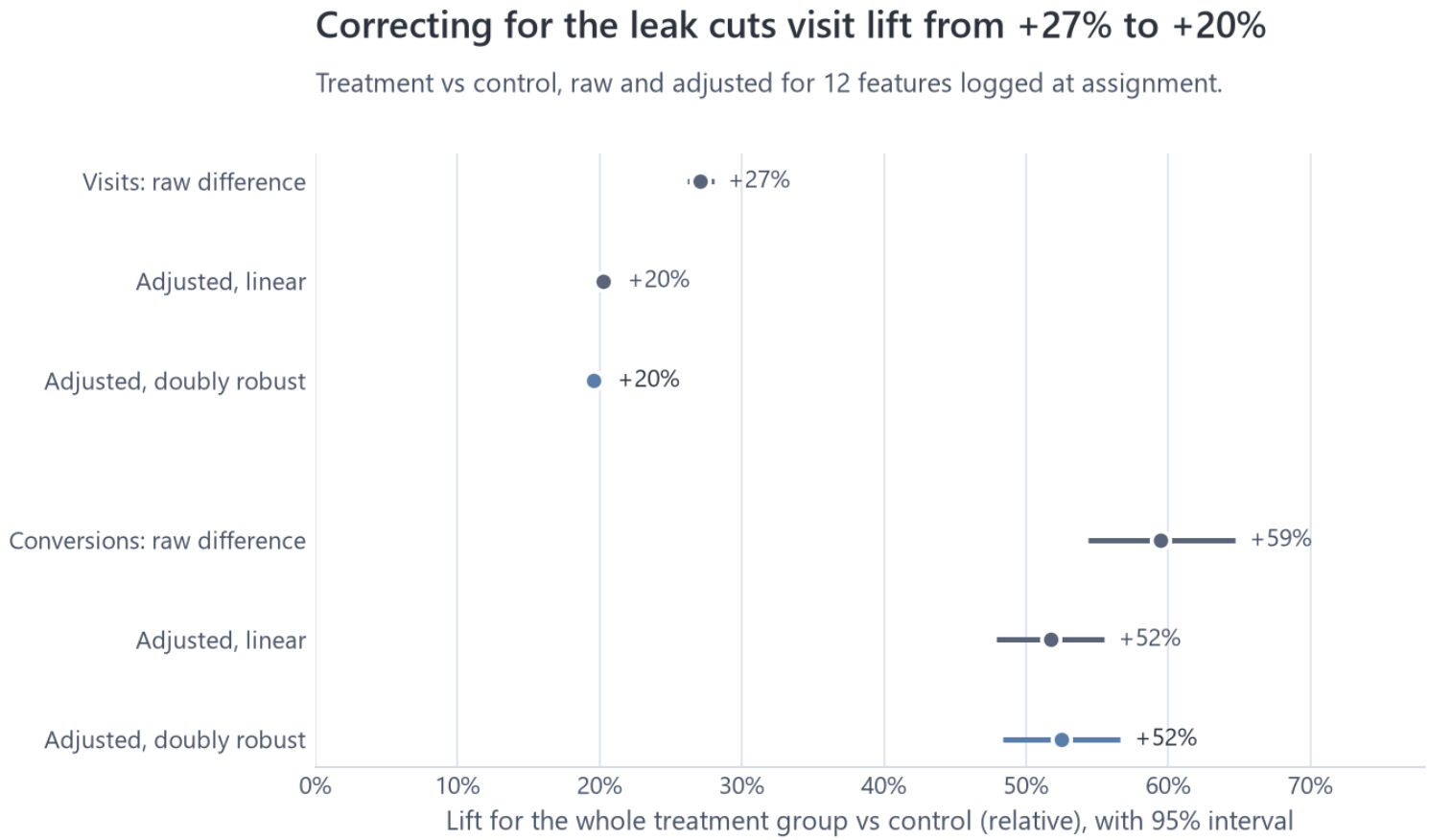
And a marketing mix model for cross-channel budgets, anchored by these experiments

Code, notebooks, dashboard and report

github.com/sonny-888

linkedin.com/in/sharath-chandra-2bb604377

Sharath Sista · Data Analyst



Source: Criteo Uplift v2.1 (Diemert et al., AdKDD 2018), randomised test on 13,979,592 users.

- Intent-to-treat with intervals; Lin and doubly robust (AIPW, cross-fitted) adjustment
- Two-stage least squares for the effect of seeing an ad
- T- and X-learner uplift graded on 2.8M held-out users
- Difference-in-differences, synthetic control, structural time series

Source: Criteo Uplift v2.1 (Diemert et al.), CC BY-NC-SA 4.0 · levels describe the sub-sampled benchmark

What this does and doesn't show

- Criteo sub-sampled the file: levels describe this benchmark; the methods transfer
- No prices or costs: money uses stated assumptions, with sensitivity tables
- Spend is proxied by ad views; real cost depends on frequency and price
- One geo dataset: liquor wholesale orders are lumpy; smoother sales detect smaller lifts